

Module: 1, Introduction to Statistics

Statistics: Introduction

- In the modern world of computers and information technology, the importance of statistics is very well recognized by all the disciplines.
- Statistics has originated as a science of statehood and found applications slowly and steadily in Agriculture, Economics, Commerce, Biology, Medicine, Industry, planning, education and so on.

Definition of Statistics

- Statistics is defined differently by different authors over a period of time
- Statistics are numerical statement of facts in any department of enquiry placed in relation to each other. - A.L. Bowley
- Statistics may be defined as the science of collection, presentation analysis and interpretation of numerical data from the logical analysis. It is clear that the definition of statistics by Croxton and Cowden is the most scientific and realistic one. According to this definition there are four stages:Collection of Data,Presentation of data,Analysis of data and Interpretation of data. - Croxton and Cowden

Data Analysis

- The data can be collected in connection with time or geographical location or in connection with time and location.
- Any statistical data can be classified under two categories depending upon the sources utilized. Primary data Secondary data.

Frequency distribution

- Frequency distribution is a series when a number of observations with similar or closely related values are put in separate bunches or groups, each group being in order of magnitude in a series.
- It is simply a table in which the data are grouped into classes and the number of cases which fall in each class are recorded.
- It shows the frequency of occurrence of different values of a single Phenomenon.

A frequency distribution is constructed for three main reasons:

- To facilitate the analysis of data.
- To estimate frequencies of the unknown population distribution from the distribution of sample data
- To facilitate the computation of various statistical measures

Raw data or Ungrouped data

The statistical data collected are generally raw data or ungrouped data.

Example 1

Let us consider the daily wages (in Rs) of 30 labourers in a factory.

800, 700, 550, 500, 600, 650, 400, 300, 800, 900, 750, 450, 350, 650, 700,
800, 820, 550, 650, 800, 600, 550, 380, 650, 750, 850, 900, 650, 450, 750.

Discrete (or) Un grouped frequency distribution

In this form of distribution, the frequency refers to discrete value. Here the data are presented in a way that exact measurement of units are clearly indicated.

1	0	3	2	1	5	6	2
2	1	0	3	4	2	1	6
3	2	1	5	3	3	2	4
2	2	3	0	2	1	4	5
3	3	4	4	1	2	4	5

Represent the data in the form of a discrete frequency distribution.

Number of Children	Frequency
0	3
1	7
2	10
3	8
4	6
5	4
6	2
Total	40

Continuous frequency distribution

In this form of distribution refers to groups of values. This becomes necessary in the case of some variables which can take any fractional value and in which case an exact measurement is not possible. Hence a discrete variable can be presented in the form of a continuous frequency distribution.

Example 2

Wage distribution of 100 employees

Weekly wages (Rs)	Number of employees
50-100	4
100-150	12
150-200	22
200-250	33
250-300	16
300-350	8
Total	100

MEASURES OF CENTRAL TENDENCY

- In the study of a population with respect to one in which we are interested we may get a large number of observations.
- It is not possible to grasp any idea about the characteristic when we look at all the observations.
- So it is better to get one number for one group. That number must be a good representative one for all the observations to give a clear picture of that characteristic.
- Such representative number can be a central value for all these observations. This central value is called a measure of central tendency or an average or a measure of location

There are five averages.

- Mean
- Geometric mean
- Harmonic mean
- Median
- Mode

Arithmetic mean or Mean

Arithmetic mean or simply the mean of a variable is defined as the sum of the observations divided by the number of observations. If the variable x assumes n values $x_1, x_2, \dots, x_n$ then the mean, $\bar{x}$ is given by

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The mean for grouped data is obtained from the following formula:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n x_i f_i$$

where x_i - the given individual data value

f_i - the frequency of individual class

N - the sum of the frequencies or total frequencies.

Note: For continuous frequency distribution, x_i — the mid-point value of individual class

Problem 1

Given the following frequency distribution, Calculate the arithmetic mean

Marks	64	63	62	61	60	59
No. of students	8	18	12	9	7	6

Solution:

x_i	f_i	$f_i x_i$
64	8	512
63	18	1134
62	12	744
61	9	549
60	7	420
59	6	354
	N=60	$\sum_{i=1}^6 f_i x_i = 3713$

$$\bar{x} = 61.88$$

Problem 2

Following is the distribution of persons according to different income groups. Calculate arithmetic mean.

<i>Income(Rs)</i>	<i>0-10</i>	<i>10-20</i>	<i>20-30</i>	<i>30-40</i>	<i>40-50</i>	<i>50-60</i>	<i>60-70</i>
<i>No. of persons</i>	<i>6</i>	<i>8</i>	<i>10</i>	<i>12</i>	<i>7</i>	<i>4</i>	<i>3</i>

Solution:

Income C.I	Number of Students (f)	Mid x	fx
0-10	6	05	030
10-20	8	15	120
20-30	10	25	250
30-40	12	35	420
40-50	7	45	315
50-60	4	55	220
60-70	3	65	195
Total	N=50		1550

$$N = 50, \sum_{i=1}^7 x_i f_i \text{ and } \bar{x} = \frac{1}{N} \sum_{i=1}^7 x_i f_i = \frac{1}{50}(1550) = 31.$$

If $x_1, x_2, \dots, x_n$ be the values of a variable x with respective weights of $w_1, w_2, \dots, w_n$ assigned to them, then

$$\text{Weighted A.M} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}.$$

Harmonic mean

Harmonic mean of a set of observations is defined as the reciprocal of the arithmetic average of the reciprocal of the given values. If $x_1, x_2, \dots, x_n$ are n observations.

$$\text{Harmonic Mean}(H.M) = \frac{n}{\sum_{i=1}^n \left(\frac{1}{x_i} \right)}$$

For frequency distribution,

$$\text{Harmonic Mean}(H.M) = \frac{N}{\sum_{i=1}^n f_i \left(\frac{1}{x_i} \right)}$$

Problem 3

Calculate Harmonic Mean

<i>Family (Rs)</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>
<i>Income</i>	<i>85</i>	<i>70</i>	<i>10</i>	<i>75</i>	<i>500</i>	<i>8</i>	<i>42</i>	<i>250</i>	<i>40</i>	<i>36</i>

ANS: 28.83

Problem 4

Calculate Harmonic Mean

<i>(x) Size of items</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>	<i>11</i>
<i>(f) Frequency</i>	<i>4</i>	<i>6</i>	<i>9</i>	<i>5</i>	<i>2</i>	<i>8</i>

ANS: 8.23

Geometric Mean

The geometric mean of a series containing n observations is the n th root of the product of the values. If $x_1, x_2, \dots, x_n$ are n observations then

$$G.M = \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n}$$

An alternative formula

$$\begin{aligned} G.M &= \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n} \\ \log(G.M) &= \log(x_1 \cdot x_2 \cdot \dots \cdot x_n)^{\left(\frac{1}{n}\right)} \\ &= \frac{1}{n} \left(\sum_{i=1}^n \frac{\log(x_i)}{n} \right) \\ G.M &= \text{Antilog} \left(\sum_{i=1}^n \frac{\log(x_i)}{n} \right) \end{aligned}$$

Problem 5

Calculate the Geometric mean

(1) 2, 4, 2, 8, 5.

(2) 50, 72, 54, 82, 93

In the case of frequency distribution

$$G.M = \text{Antilog} \left(\frac{\sum_{i=1}^n f_i \log(x_i)}{\sum_{i=1}^n f_i} \right)$$

Example 3

Calculate G.M

Weight	130	135	140	145	146	148	149
(<i>f</i>)	3	4	6	6	3	5	2

Median

The median is that value of the variate which divides the group into two equal parts, one part comprising all values greater, and the other, all values less than median.

- **Ungrouped data:**

Arrange the given values in the increasing or decreasing order.

If the number of values are odd, median is the middle value.

If the number of values are even, median is the mean of middle two values.

Problem 6

Find median for the following data

- 25, 18, 27, 10, 8, 30, 42, 20, 53.
- 8, 20, 50, 25, 15, 30.

Problem 7

The following table represents the marks obtained by a batch of 10 students in certain class tests in Chemistry and Physics.

<i>Students ID</i>	<i>Marks in Chemistry</i>	<i>Marks in Physics</i>
1	28	57
2	55	45
3	53	24
4	32	31
5	30	25
6	60	84
7	47	43
8	46	80
9	35	32
10	52	72

Indicate in which subject is the level of knowledge higher?

Solution

The marks in the two subjects are first arranged in increasing order as follows:

Marks in Chemistry: 28, 30, 32, 35, 46, 47, 52, 53, 55, 60.

Marks in Physics: 24, 25, 31, 32, 43, 45, 57, 72, 80, 84.

Here $n = 10$.

Median(Chemistry) is the mean of the middle two values (ie) mean of (46,47). That is $(46 + 47)/2 = 46.5$ is the median(Chemistry).

Median(Physics) is the mean of the middle two values (ie) mean of (43,45). That is $(43 + 45)/2 = 44$ is the median(Physics).

Therefore the level of knowledge in Chemistry is higher than that in Physics.

Cumulative frequency (cf)

Cumulative frequency of each class is the **sum of the frequency of the class and the frequencies of the pervious classes**, ie adding the frequencies successively, so that the last cumulative frequency gives the total number of items.

Median-Grouped Data: Discrete frequency

Procedure for calculating median is given as follows:

- Find cumulative frequencies.
- Find $\frac{N}{2}$
- Look for the cumulative frequencies the value just greater than $\frac{N}{2}$
- Then the corresponding value of x is median.

Problem 8

Obtain the median for the following frequency distribution.

x	1	2	3	4	5	6	7	8	9
(f)	8	10	11	16	20	25	15	9	6

ANS: 5

Median-Grouped Data:Continuous frequency

Procedure for calculating median is given as follows:

- Find cumulative frequencies Find $\frac{N}{2}$
- See in the cumulative frequencies the value just greater than $\frac{N}{2}$
- Then the corresponding class interval is called Median class.
- Then apply the formula:

$$Median = l + \frac{h}{f} \left(\frac{N}{2} - cf \right)$$

Where l = Lower limit of the median class

cf = cumulative frequency preceding the median class

h = width of the median class

f = frequency in the median class. N = Total frequency.

Problem 9

Following is the distribution of persons according to different income groups. Calculate median.

<i>Income(Rs)</i>	<i>20-30</i>	<i>30-40</i>	<i>40-50</i>	<i>50-60</i>	<i>60-70</i>
<i>Number of persons</i>	<i>3</i>	<i>5</i>	<i>20</i>	<i>10</i>	<i>5</i>

Solution

Here $N = 43$. Then $N/2 = 21.5$

Income c.I	Number of persons (f)	cf
20-30	3	3
30-40	5	8
40-50	20	28
50-60	10	38
60-70	5	43

Therefore, in the cumulative frequencies the value just greater than $N/2 = 21.5$ is 28. Then the corresponding class interval (ie) median class is 40 – 50.

Then Median = $l + \frac{h}{f} \left(\frac{N}{2} - cf \right)$, where

Lower limit of the median class $l = 40$ Cumulative frequency preceding the median class $cf = 8$ Width of the median class $h = 10$ Frequency in the median class $f = 20$ and Total frequency $N = 43$.

$$\begin{aligned}
 \text{Median} &= l + \frac{h}{f} \left(\frac{N}{2} - cf \right) \\
 &= 40 + 10 \left(\frac{\frac{43}{2} - 8}{20} \right) \\
 &= 40 + \frac{13.5}{2} = 46.75.
 \end{aligned}$$

Problem 10

*The following data attained from a garden records of certain period
Calculate the median weight of the apple.*

<i>Weight in grams</i>	<i>Number of apples</i>
410-420	14
420-430	20
430-440	42
440-450	54
450-460	45
460-470	18
470-480	7

ANS: 444.45

Mode

- Mode is defined as the value which occurs most frequently in a data set. In other words, mode is the value of the variable which is predominant in the series.
- It is the value around which the items or observations tend to be most heavily concentrated.
- **Un grouped Data:**

The mode is defined as the value which occurs frequently in a data set.

Problem 11

A teacher was conducted a test for test 20 students and their marks are recorded 90, 70, 50, 30, 40, 86, 65, 73, 68, 90, 90, 10, 73, 25, 35, 88, 67, 80, 74, 46. Find the mode value of marks.

Problem 12

A doctor who checked 9 patient's sugar level is recorded 80, 112, 110, 115, 124, 130, 100, 90, 150, 180. Find the mode value of the sugar level.

Problem 12

A doctor who checked 9 patient's sugar level is recorded 80, 112, 110, 115, 124, 130, 100, 90, 150, 180. Find the mode value of the sugar level.

Solution: Since every value appear only once in the data set, it follows that there is no mode.

Problem 13

Compute mode value for the following observations 2, 7, 10, 12, 10, 19, 2, 11, 3, 12.

Solution: Here, the values 2, 10 and 12 are appear twice in the data set. Therefore, the modes are 2, 10 and 12.

• It is clear that mode may not exist or mode may not be unique

Mode: Discrete series

For discrete frequency distribution, mode is the value of the variable corresponding to the maximum frequency.

Problem 14

Calculate the mode from the following data

<i>Days of Confinement</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>
<i>Number of patients</i>	<i>4</i>	<i>6</i>	<i>7</i>	<i>5</i>	<i>2</i>

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For discrete frequency distribution, mode is the value of the variable corresponding to the maximum frequency.

Problem 14

Calculate the mode from the following data

<i>Days of Confinement</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>
<i>Number of patients</i>	<i>4</i>	<i>6</i>	<i>7</i>	<i>5</i>	<i>2</i>

Solution: Here, 7 is the maximum frequency, hence the value of x corresponding to 7 is 8. Therefore, the mode is 8.

Determination of Modal class

For a frequency distribution modal class corresponds to the class with maximum frequency. But in any one of the following cases that is not easily possible.

- If the maximum frequency is repeated.
- If the maximum frequency occurs in the beginning or at the end of the distribution.
- If there are irregularities in the distribution, the modal class is determined by the method of grouping.

We prepare a grouping table with 6 columns.

- (1) In column I, we write down the given frequencies.
- (2) Column II is obtained by combining the frequencies two by two.
- (3) Leave the first frequency and combine the remaining frequencies two by two and write in column III.
- (4) Column IV is obtained by combining the frequencies three by three.
- (5) Leave the first frequency and combine the remaining frequencies three by three and write in column V.
- (6) Leave the first and second frequencies and combine the remaining frequencies three by three and write in column VI.

Mark the highest frequency in each column. Then form an analysis table to find the modal class. Use the mode formula to calculate the modal value.

Problem 15

Find the Mode of the following frequency distribution.

<i>Size(x)</i>	1	2	3	4	5	6	7	8	9	10	11	12
<i>Frequency(f)</i>	3	8	15	23	35	40	32	28	20	45	14	6

Problem 15

Find the Mode of the following frequency distribution.

<i>Size(x)</i>	1	2	3	4	5	6	7	8	9	10	11	12
<i>Frequency(f)</i>	3	8	15	23	35	40	32	28	20	45	14	6

Solution:

Here we can see that the distribution is not regular the frequencies are increasing up to 40 and then decrease but the frequency 45 after 20 does not seem to be consistent with the distribution.

Here we can't say that since the maximum frequency is 45, the mode is 10. Here we shall locate the mode by the method of grouping.

Solution continued

Size (x)	Frequency					
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
1	3	11	23	26	46	73
2	8					
3	15					
4	23	38	58	98	107	100
5	35					
6	40					
7	32	60	72	80	93	79
8	28					
9	20					
10	45	65	59	65		
11	14					
12	6					

In the above table, the maximum frequency in each column is given in bold. To find mode we form the following table.

Solution continued

Column Number (A)	Maximum Frequency (B)	Values of x maximum frequencies in (B) (C)
(i)	45	10
(ii)	75	5,6
(iii)	72	6,7
(iv)	98	4,5,6
(v)	107	5,6,7
(vi)	100	6,7,8

On examining the values in column (C) above, we find that the value of 6 is repeated the maximum number of times and hence the value of mode is 6 and not 10 which is an irregular item.

Mode: Continuous series

- The mode or modal value of the continuous distribution is computed by the formula.

$$Mode = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h.$$

- Where,
 l is the lower limit of the modal class.
 f_1 is the frequency of the modal class.
 f_0 is the frequency of the class preceding the modal class.
 f_2 is the frequency of the class succeeding the modal class.
 h is the width of the class limit.

Remark 16

- If $2f_1 - f_0 - f_2$ comes out to be zero, then mode is obtained by the following formula taking absolute differences.

- $$\text{Mode} = l + \left(\frac{f_1 - f_0}{|f_1 - f_0| + |f_1 - f_2|} \right) \times h.$$

Problem 17

The following data relates to the daily income of families in an urban area. Find the modal income of the families.

<i>Income(Rs)</i>	<i>Number of persons</i>
<i>0-100</i>	<i>5</i>
<i>100-200</i>	<i>7</i>
<i>200-300</i>	<i>12</i>
<i>300-400</i>	<i>18</i>
<i>400-500</i>	<i>16</i>
<i>500-600</i>	<i>10</i>
<i>600-700</i>	<i>5</i>

Solution

- Here the maximum frequency is 18, therefore the modal class is 300-400.
- The lower limit of the modal class $l = 300$
- The frequency of the modal class $f_1 = 18$
- The frequency of the class preceding the modal class $f_0 = 12$
- The frequency of the class succeeding the modal class $f_2 = 16$
- The width of the class limit $h = 100$.
- Mode $= l + \left(\frac{f_1 - f_2}{2f_1 - f_0 - f_2} \right) \times h = 300 + \left(\frac{18 - 16}{2(18) - 12 - 16} \right) \times 100 = 300 + 0.75(100) = 375$.

Problem 18

Calculate mode for the following frequency distribution:

<i>Income(Rs)</i>	<i>Number of persons</i>
<i>0-5</i>	<i>9</i>
<i>5-10</i>	<i>12</i>
<i>10-15</i>	<i>15</i>
<i>15-20</i>	<i>16</i>
<i>20-25</i>	<i>17</i>
<i>25-30</i>	<i>15</i>
<i>30-35</i>	<i>10</i>
<i>35-40</i>	<i>13</i>

Solution

Since there are irregularities in the distribution, because the frequency values are increasing order 9,12,15,16,17 then gradually decreasing 15,10 but the next value 13 is increasing from the previous value 10. Therefore, use grouping method to determine the Modal class.

Solution continued

Class	I	II	III	IV	V	VI
0-5	9	21				
5-10	12		27	36		
10-15	15	31			43	
15-20	16					48
20-25	17	32	33	48		
25-30	15		25		42	
30-35	10					38
35-40	13	23				

Solution continued

Class	I	II	III	IV	V	VI	Total
0-5							
5-10					1		1
10-15					1	1	2
15-20			1	1	1	1	4
20-25	1	1	1	1		1	5
25-30		1		1			2
30-35							
35-40							

The maximum number of 1's appear in the class interval is 20-25.
Therefore the Modal class is 20-25.

Solution continued

- The lower limit of the modal class $l = 20$
- The frequency of the modal class $f_1 = 17$
- The frequency of the class preceding the modal class $f_0 = 16$
- The frequency of the class succeeding the modal class $f_2 = 15$
- The width of the class limit $h = 5$.

Mode

$$= l + \left(\frac{f_1 - f_2}{2f_1 - f_0 - f_2} \right) \times h = 20 + \left(\frac{17 - 16}{2(17) - 16 - 15} \right) \times 5 = 20 + \frac{1}{3} \times 5 = 21.65.$$

Problem 19

Find the mean, median and mode for the following distribution.

C	130-134	135-139	140-144	145-149	150-154	155-159	160-
(f)	5	15	28	24	17	10	1

c.f	freq	c.f
129.5 - 134.5	5	5
134.5 - 139.5	15	20
139.5 - 144.5	28	48
144.5 - 149.5	24	72
149.5 - 154.5	17	89
154.5 - 159.5	10	99
159.5 - 164.5	1	100

$$\boxed{\frac{N}{2} = 50}$$

$$\text{Mean} = \frac{\sum x_i f_i}{\sum f_i} = \frac{132.5 \times 5 + 137.5 \times 15 + 142.5 \times 28 + \dots + 162.5 \times 1}{100}$$

$$= 145.85 \sim 146 \quad (\text{or}) = A + \frac{\sum f d}{N} \cdot h = 145.35$$

$$\text{Median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

$$= 144.5 + \frac{5}{24} (50 - 48)$$

$$= 144.92 \sim 145$$

$$\text{Mode} = l + \frac{h(f_0 - f_1)}{2f_0 - f_1 - f_2}$$

$$= 139.5 + \frac{5(28 - 15)}{2 \times 28 - 15 - 24}$$

Problem 20

Find the mean, median and mode for the following distribution.

(1)

Class	500-700	700-900	900-1100	1100-1300	1300-1500
(f)	5	11	26	10	8

(2)

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
(f)	5	8	7	12	28	20	10	10

(3)

Class	11-20	21-30	31-40	41-50	51-60
(f)	12	10	15	16	20

Problem 21

The median and mode of the following wage distribution are known to be Rs. 33.5 and Rs. 34 respectively. Find the values of f_3 , f_4 and f_5 .

Wages	0-10	10-20	20-30	30-40	40-50	50-60	60-70	Total
(f)	4	16	f_3	f_4	f_5	6	4	230

Measures of Dispersion

The various measures of averages give a single number as the representative of the whole data. But they do not give how the observations are scattered about the average.

Example 4

Two distributions giving the weekly wages of 200 persons may have the same mean value, say Rs. 100. In one distribution, most of the observations may be centered around the mean value 100, a few others away from 100. In another distribution a large number of observations may be above 150 and another set of large number of observations may be below 50 and only a few between 50 and 100 and still mean may be 100.

From this example,

- These two distributions with the same mean are not identical.
- In one, the items are nearer to the mean and in the other they are spread away from the mean.
- Two distributions may also have same median. But the deviations of the observations from the median may be different type in the two distributions.
- To study this aspect of distributions, another characteristic called dispersion.

Measures of Dispersion

There are two kind of measures of dispersion

- Absolute measure of dispersion
- Relative measure of dispersion

Absolute measure of dispersion: Absolute measure of dispersion indicates the amount of variations in a set of values in terms of units of observation.

Example

When rainfalls on differ days are available in mm, any absolute measure of dispersion gives the variation rainfall in mm.

Relative measure of dispersion

Relative measures of dispersion are free from the units of measurements of the observations. They are pure numbers. They are used to compare the variation in two or more sets, which are having different units of measurement of observation.

Absolute measure

- Range
- Quartile deviation
- Mean deviation
- Standard deviation

Relative measure

- Co-efficient of Range
- Co-efficient of Quartile deviation
- Co-efficient of Mean deviation
- Co-efficient of variation

Range

Range = $L - S$, where L -Largest value, S -Smallest value.

Co-efficient of Range

Co-efficient of Range = $\frac{L-S}{L+S}$, where L -Largest value, S -Smallest value.

Find L and S for Continuous Series**Method-I**

L -Upper boundary of the highest class

S -Lower boundary of the lowest class

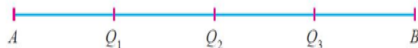
Method-II

L -Mid value of the highest class

S -Mid value of the lowest

Quartile deviation

The quartiles divided the distributions in four equal parts.



- The first quartile(Q_1) marks off the first one-fourth
- The second quartile divides the distribution into two halves (median)(Q_2).
- The third quartile(Q_3) marks off the three-fourth.
- Arrange the given data in the increasing order
- Use the formula for $Q_1 = (\frac{n+1}{4})^{th}$ value
- Use the formula for $Q_2 = (\frac{n+1}{2})^{th}$ value
- Use the formula for $Q_3 = (\frac{3(n+1)}{4})^{th}$ value

Note: If Quartile position is not an integer, use the equation to find the quartile value.

Quartile value = Approximate quartile value + fraction part of quartile position $\times$ (consecutive value after quartile position - approximate quartile value).

Example 5

Find the range of weights of seven students from the following 27, 30, 35, 36, 38, 40, 43.

Example 6

Compute quartiles for the data 25, 18, 30, 8, 15, 5, 10, 35, 40, 45

Quartiles: Grouped data-Discrete series

- Find Cumulative frequencies(c.f)
- Find $\frac{N}{4}$, where N is the total frequency
- See in the cumulative frequencies, the value just greater than $\frac{N}{4}$
- Then the corresponding value of x is Q_1 .
- Find $\frac{3(N)}{4}$
- See in the cumulative frequencies, the value just greater than $\frac{3(N)}{4}$
- Then the corresponding value of x is Q_3 .

Problem 22

Compute quartiles for the data given below

x	f
5	4
8	3
12	2
15	4
19	5
24	2
30	4

Solution.

x	f	cf
5	4	4
8	3	7
12	2	9
15	4	13
19	5	18
24	2	20
30	4	24
	N=24	

Here $N = 24$ and $\frac{N}{4} = 6$. The cf value just greater than is 7 and the corresponding x value is 8. So, $Q_1 = 8$.

Here we have $Q_2 = 15$.

Also, $\frac{3N}{4} = 18$. The cf value just greater than 18 is 20. The corresponding x value is 24. Thus $Q_3 = 24$.

Quartiles: Grouped data-Continuous series

- Find Cumulative frequencies(c.f)
- Find $\frac{N}{4}$, where N is the total frequency
- See in the cumulative frequencies, the value just greater than $\frac{N}{4}$, then the corresponding class interval is called first quartile class.
- Find $\frac{3N}{4}$ and see in the cumulative frequencies, the value just greater than $\frac{3N}{4}$, then the corresponding class interval is called third quartile class.
- Then apply the respective

$$Q_1 = l_1 + \frac{h_1}{f_1} \left(\frac{N}{4} - cf_1 \right)$$

$$Q_3 = l_3 + \frac{h_3}{f_3} \left(\frac{3N}{4} - cf_3 \right),$$

where

- l_1 = Lower limit of the first quartile class
- l_3 = Lower limit of the third quartile class
- cf_1 = cumulative frequency preceding the first quartile class
- cf_3 = cumulative frequency preceding the third quartile class
- h_1 = width of the first quartile class
- h_3 = width of the third quartile class
- f_1 = frequency of the first quartile class
- f_3 = frequency of the third quartile class
- N = Total frequency.

Problem 23

The following series relates to the marks secured by students in an examination. Compute quartiles.

Marks	Number of students
0-10	11
10-20	18
20-30	25
30-40	28
40-50	30
50-60	33
60-70	22
70-80	15
80-90	12
90-100	10

Quartile deviation(Q.D)

Quartile deviation(Q.D)

Quartile deviation is half of the difference between the first and third quartiles

$$\text{Quartile deviation}(Q.D) = \frac{Q_3 - Q_1}{2}$$

$$\text{Co-efficient of } (Q.D) = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Problem 24

Calculate quartile deviation and its coefficients of A's monthly earning for a year

<i>Months</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>
<i>Earning</i>	<i>239</i>	<i>250</i>	<i>251</i>	<i>251</i>	<i>277</i>	<i>258</i>	<i>260</i>	<i>261</i>	<i>262</i>

<i>10</i>	<i>11</i>	<i>12</i>
<i>262</i>	<i>273</i>	<i>275</i>

Problem 25

Calculate the range and quartile deviation of wages

<i>Wages</i>	<i>Employees</i>
<i>30-32</i>	<i>12</i>
<i>32-34</i>	<i>18</i>
<i>34-36</i>	<i>16</i>
<i>36-38</i>	<i>14</i>
<i>38-40</i>	<i>12</i>
<i>40-42</i>	<i>8</i>
<i>42-44</i>	<i>6</i>

Solution

X	f	cf
30-32	12	12
32-34 Q_1	18	30
34-36	16	46
36-38	14	60
38-40 Q_2	12	72
40-42	8	80
42-44	6	86

Here $N = 86$, $\frac{N}{4} = 21.5$ $\frac{3N}{4} = 64.5$

* The cf is just greater than $\frac{N}{4} = 21.5$ is 30.
 $\therefore$ Corresponding class contains Q_1
 $\therefore Q_1 = l + \frac{h}{f} \left(\frac{N}{4} - c \right)$

$$= 32 + \frac{2}{18} (21.5 - 12)$$

$$= 33.06$$

* The cf is just greater than $\frac{3N}{4}$ is 72.

Range
38-40

$$Q_3 = l + \frac{h}{f} \left(\frac{3N}{4} - c \right)$$

$$= 38.75$$

$$\therefore Q.D = \frac{Q_3 - Q_1}{2} = 2.85$$

Problem 26

Find Q.D

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of students	6	5	8	15	7	6	3

Deciles

These are the values, which divide the total number of observation into 10 equal parts. Therefore, there are 9 deciles, $D_1, D_2, \dots, D_9$.

Deciles for grouped data: Discrete frequency

- Find Cumulative frequencies(c.f)
- Find $i \times \frac{N}{10}$, where N is the total frequency
- See in the cumulative frequencies, the value just greater than $i \times \frac{N}{10}$,
- Then the corresponding value of x is D_i

Deciles for grouped data: Continuous frequency

- Find Cumulative frequencies(c.f)
- Find $i \times \frac{N}{10}$, where N is the total frequency
- See in the cumulative frequencies, the value just greater than $i \times \frac{N}{10}$,
- Then the corresponding class interval is called decciles class.
- Then apply the respective formula

$$D_i = l_i + \frac{h_i}{f_i} \left(i \times \frac{N}{10} - cf_i \right)$$

Where

l_i = Lower limit of the decciles class

cf_i = Cumulative frequency preceding the decciles class

h_i = Width of the decciles class

fi = Frequency of the decciles class

N =Total frequency.

Problem 27

The following series relates to the marks secured by students in an examination. Calculate D_3 and D_7 .

Marks	Number of students
0-10	5
10-20	7
20-30	12
30-40	16
40-50	10
50-60	8
60-70	4

Percentiles

- The percentile values divide the distribution into 100 parts each containing 1 percent of the cases. It is denoted by P_k .
- The percentile P_k is that value of the variable up to which lie exactly $k\%$ of the total number of observations.
- Relationship $P_{25} = Q_1$, $P_{50} = D_5 = Q_2 = \text{median}$ and $P_{75} = Q_3$.

Percentiles for grouped data-Discrete series

- Find Cumulative frequencies(c.f)
- Find $i \times \frac{N}{100}$, where N is the total frequency
- See in the cumulative frequencies, the value just greater than 100
- Then the corresponding value of x is P_i

Percentiles for Grouped data-Continuous series

- Find Cumulative frequencies(c.f)
- Find $i \times \frac{N}{100}$, where N is the total frequency
- See in the cumulative frequencies, the value just greater than $i \times \frac{N}{100}$.
- Then the corresponding class interval is called percentiles class.
- Then apply the respective formula

$$P_i = l_i + \frac{h_i}{f_i} \left(i \times \frac{N}{100} - cf_i \right)$$

where

l_i = Lower limit of the percentiles class

cf_i = Cumulative frequency preceding the percentiles class

h_i = Width of the percentiles class

f_i = Frequency of the percentiles class

N = Total frequency.

Problem 28

The following series relates to the marks secured by students in an examination. Calculate P_{53} .

Marks	Number of students
0-5	5
5-10	8
10-15	12
15-20	16
20-25	20
25-30	10
30-35	4
35-40	4
40-45	3

Mean Deviation

- Mean deviation calculated by any measure of central tendency is an absolute measure.
- For the purpose of comparing variation among different series, a relative mean deviation is required.
- The relative mean deviation is obtained by dividing the mean deviation by the average used for calculating mean deviation.

$$\text{Coefficient of mean deviation} = \frac{\text{Mean deviation}}{\text{Mean or Median or Mode}}$$

Mean deviation-Ungrouped data

- Calculate the average(mean, median or mode) of the series.
- Take the deviations of items from average ignoring signs and denote these deviations by $|D|$.
- Compute the total of these deviations, i.e., $\sum |D|$
- Divide this total obtained by the number of items.

Example 7

Calculate mean deviation from mean and median for the following data: 100, 150, 200, 250, 360, 490, 500, 600, 671. Also, calculate coefficients of M.D.

Solution

Wages	$ D = x - \bar{x}$	$ D = x - \text{median}$
100	269	260
150	219	210
200	169	160
250	119	110
360	9	0
490	121	130
500	131	140
600	231	240
671	302	311
$\sum x = 3321$	$\sum D = 1570$	$\sum D = 1561$

- Mean $\bar{x} = \frac{\sum x}{n} = \frac{3321}{9} = 369$. Median = 360.
- Mean deviation(from Mean) $= \frac{\sum |D|}{n} = \frac{1570}{9} = 174.44$.
- Coefficient of M.D(from mean) $= \frac{M.D}{\bar{x}} = \frac{174.44}{369} = 0.4727$.
- Mean deviation(from Median) $= \frac{\sum |D|}{n} = \frac{1561}{9} = 173.44$.
- Coefficient of M.D(from Median) $= \frac{M.D}{\bar{x}} = \frac{173.44}{360} = 0.4817$.

Mean deviation-Grouped data:Discrete Series

- Calculate the average(mean, median or mode) of the series.
- Find out the deviation of the variable values from the average, ignoring signs and denote these deviations by $|D|$.
- Compute the total of these deviations, i.e., $\sum f|D|$
- Divide this total obtained by the total frequency N . i.e.,

$$M.D = \frac{\sum f|D|}{N}$$

Problem 29

Calculate Mean deviation from the following table.

x	2	4	6	8	10
f	1	4	6	4	1

Solution

x	f	fx	$ D = x - \bar{x} $	$f D $
2	1	2	4	4
4	4	16	2	8
6	6	36	0	8
8	4	32	2	8
10	1	10	4	4
$N = \sum f = 16$		$\sum fx = 96$		$\sum f D = 24$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{96}{16} = 6.$$

$$\text{Mean deviation} = \frac{\sum f|D|}{\sum f} = \frac{24}{16} = 1.5.$$

Problem 30

Calculate mean deviation from mean and median for the following data: Also, calculate coefficients of M.D

<i>Height in cms</i>	<i>Number of Persons</i>
158	15
159	20
160	32
161	35
162	33
163	22
164	10
165	20
166	8

Mean deviation-Grouped data:continuous Series

- Calculate the average (mean, median or mode) of the series.
- Compute $|D| = |Mid - Average|$, where Mid is the mid point of class interval.
- Compute $\sum f|D|$
- Divide this total obtained by the total frequency N .

$$M.D = \frac{\sum f|D|}{N}$$

Problem 31

Calculate mean deviation from mean and median for the following data: Also, calculate coefficients of M.D.

Marks	Number of students
0-10	20
10-20	25
20-30	32
30-40	40
40-50	42
50-60	35
60-70	10
70-80	8

Solution

- To find Mean and Median.

Mean $\bar{x} = \frac{\sum fMid}{N}$, where Mid is the mid point of the class interval and N is the total frequency.

Therefore, $\bar{x} = 36.5$

- To find Median

Now, $\frac{N}{2} = \frac{212}{2} = 106$

Therefore, Median class is 30-40

$$\text{Median } Md = l + h \left(\frac{\frac{N}{2} - cf}{f} \right)$$

Here $l = 30$, $cf = 77$, $f = 40$, $h = 10$.

Hence Median $Md = 37.25$.

Here $|D_1| = |Mid - \bar{x}|$ and $|D_2| = |Mid - Md|$, $\bar{x} = 36.5$ and $Md = 37.25$.

Class	Mid	f	fMid	c.f	$ D_1 $	$f D_1 $	$ D_2 $	$f D_2 $
0-10	5	20	100	20	31.5	630	32.5	645
10-20	15	25	375	45	21.5	537.5	22.25	556.25
20-30	25	32	800	77	11.5	368	12.25	392
30-40	35	40	1400	117	1.5	60	2.25	90
40-50	45	42	1890	159	8.5	357	7.75	325.5
50-60	55	35	1925	194	18.5	647.5	17.75	621.25
60-70	65	10	650	204	28.5	285	27.75	277.5
70-80	75	8	600	212	38.5	308	37.75	302
Total		212	7740			3192.5		3209.5

$$\text{Mean deviation from mean } M.D_1 = \frac{\sum f|D_1|}{N} = \frac{3192.5}{212} = 15.06$$

$$\text{Coefficient of mean deviation from mean} = \frac{\text{Mean deviation}}{\text{Mean}} = \frac{15.06}{36.5} = 0.41$$

$$\text{Mean deviation from median } M.D_2 = \frac{\sum f|D_2|}{N} = \frac{3209.5}{212} = 15.14$$

Coefficient of mean deviation from

$$\text{Median} = \frac{\text{Mean deviation}}{\text{Median}} = \frac{15.14}{37.25} = 0.41$$